

Socratic-OT: A Grounded Multimodal Socratic Tutor for Anatomy and Neuroscience Learning in Rehabilitation Science

Vidhyadhari Bandaru

Department of Computer Science
and Engineering
University at Buffalo
Buffalo, NY, USA
bandaru7@buffalo.edu

Richie M Ilavarapu

Department of Computer Science
and Engineering
University at Buffalo
Buffalo, NY, USA
richiemo@buffalo.edu

Abstract

We present **Socratic-OT**, a grounded multimodal tutoring system for occupational-therapy education, with emphasis on anatomy and neuroscience. The system addresses a core problem from the project brief: standard AI assistants answer too quickly, weakening guided recall and clinical reasoning. Our implementation therefore follows a strict *tutor-not-teller* policy. It retrieves only faculty-approved content, masks the target answer during the first two tutoring turns, escalates from broad hint to stronger clue, supports diagram-based tutoring, maintains session-only memory of repeated weaknesses, and ends with a lightweight mastery check. The knowledge layer combines a text corpus derived from OpenStax *Anatomy & Physiology 2e* with a curated image set of labeled anatomy and neuroscience diagrams linked through compact metadata. At the current milestone, the project includes 70 relevant textbook sections, 168 retrieval-ready text chunks, and 6 image metadata records.

1 Introduction

Large language models have made conversational tutoring widely accessible, but general-purpose assistants often provide direct answers immediately, offer weak pedagogical control, and may drift away from trusted educational sources. In occupational therapy education this is especially problematic because gross anatomy and neuroscience are bottleneck courses with high content density, strong dependence on 3D visual relationships, and a need for conceptual understanding rather than simple answer recall.

Our implementation is designed as a *grounded Socratic tutor* rather than a standard question-answering bot. The system uses retrieval for groundedness, a turn-aware masking policy for pedagogy, multimodal support for diagrams, session

memory for adaptation, and assessment for mastery checking. Although the current domain is anatomy and neuroscience for rehabilitation science, the architecture is intended to be reusable across subjects by replacing the underlying knowledge base.

2 Related Work and Technical Motivation

Our design draws on prior work in grounded tutoring, Socratic pedagogy, multimodal reasoning, and RAG evaluation.

A grounded educational assistant benefits from retrieval over approved sources rather than unrestricted open-domain generation. This motivates our use of OpenStax as the textual foundation and ChromaDB as the retrieval store (OpenStax, 2022; Chroma, 2025). For retrieval quality and groundedness evaluation, we adopt RAGAS because it provides faithfulness- and relevance-oriented metrics for RAG systems (Es et al., 2024; Ragas, 2025).

The pedagogical motivation comes from systems that treat tutoring as guided reasoning rather than answer delivery. SocraticLM is especially relevant because it frames tutoring as a thought-provoking process rather than generic QA (Liu et al., 2024). Khanmigo is also a useful design reference because it explicitly promotes tutor-not-teller behavior and guided learning over direct answers (Khan Academy, 2025).

For diagram tutoring, our multimodal path is informed by LLaVA and related visual instruction tuning work, which show that vision-language systems can support image-grounded dialogue over visual inputs (Liu et al., 2023a,b). We also view tools such as Articulate 360 and BoodleBox as contrast points: they demonstrate the value of structured educational flows and custom grounded bots, but they do not natively enforce the retrieval-plus-masking tutoring constraint required by this project (Articu-

Asset	Status	Current Count
OpenStax A&P 2e PDF	downloaded and processed	1 file
Relevant text sections	selected and cleaned	70
Retrieval-ready text chunks	generated	168
Labeled diagrams	curated	6 / 20–30 target
Image metadata records	structured	6

Table 1: Current knowledge-base construction status at Milestone 1.

late Community, 2024; BoodleBox, 2025a,b).

3 Project Objectives

The system is designed around four core objectives:

- enforce a Socratic tutor-not-teller strategy rather than direct answer delivery;
- keep all responses grounded in approved educational content;
- support both text-based and image-based tutoring;
- evaluate the system for grounding, leakage control, multimodal quality, and pedagogical usefulness.

4 Data Plan and Knowledge Base

The knowledge layer contains both textual and visual educational material. The current implementation focuses on anatomy and neuroscience concepts most relevant to occupational therapy preparation.

4.1 Text Knowledge Base

The text knowledge base uses OpenStax *Anatomy & Physiology 2e* as the approved source corpus (OpenStax, 2022). The construction pipeline is section-aware and retrieval-oriented:

1. extract full text from the textbook PDF;
2. detect section boundaries and generate section-level files;
3. clean page markers, duplicated fragments, and formatting artifacts;
4. assign topic labels such as *muscle* and *nervous system*;
5. chunk each cleaned section into paragraph groups of about 220 words with about 40 words of overlap;
6. store each chunk with metadata including source identifier, chapter, section title, topic label, and keyword summary.

This pipeline currently yields **70 relevant sections** and **168 retrieval-ready chunks**. These chunk-level units are the entries that will later be embedded with all-MiniLM-L6-v2 and indexed in ChromaDB (Sentence-Transformers, 2025; Chroma, 2025).

Example chunk evidence. Representative retrieval-ready chunks include CH0166–CH0168 from Section 19.2 *Cardiac Muscle and Electrical Activity*. These chunks capture coherent passages on the prepotential of the SA node, depolarization and repolarization dynamics, and the physiological role of calcium ions in cardiac contractile cells.

4.2 Image Knowledge Base

The visual knowledge base consists of approved anatomy and neuroscience diagrams paired with compact metadata records. Each record contains an image identifier, file name, topic, structure name, aliases, body region, and an optional functional cue. At the current milestone, the image knowledge base contains **6 diagrams** and **6 metadata records**, including neuron structure, brain lobes, nervous-system overview, brachial plexus, spinal cord section, and skeletal muscle fiber organization.

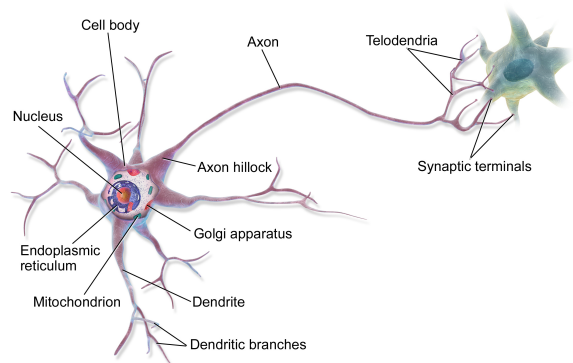


Figure 1: Neuron structure diagram used in the multi-modal tutoring path.

Field	Value
Image ID	IMG001
File name	IMG001_neuron_structure.PNG
Topic	nervous system
Structure name	Neuron
Aliases	nerve cell; neuron structure
Body region	nervous system
Function	basic neural signaling structure

Table 2: Example image metadata used for multimodal retrieval and tutoring.

5 Solution Architecture

The runtime system is organized into five layers:

1. **Knowledge layer:** textbook chunks, image files, and image metadata;
2. **Routing layer:** input classification for text, image, or follow-up messages;
3. **Retrieval layer:** semantic chunk retrieval plus metadata-guided image-to-text linking;
4. **Tutoring-policy layer:** answer masking, escalation logic, and session-memory updates;
5. **Generation layer:** text or multimodal model calls for the final grounded response.

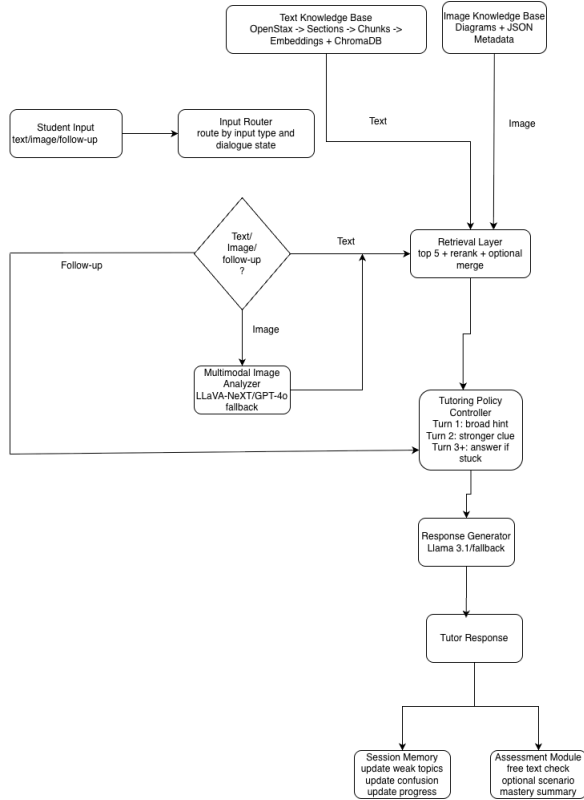


Figure 2: Socratic-OT runtime architecture. Text and image inputs share grounded retrieval, tutoring-policy control, session memory, and assessment.

5.1 Retriever

Given a student query q , the retriever computes an embedding using all-MiniLM-L6-v2 and retrieves the top- k chunk candidates from ChromaDB by semantic similarity, where $k = 5$ in the current design. Metadata-aware reranking then prioritizes chunks whose topic and section labels better align with the inferred query intent. When multiple adjacent chunks originate from the same section and jointly improve coherence, they may be merged before final prompting.

5.2 Tutoring Policy Controller

The tutoring-policy layer identifies the likely target concept internally, enforces answer masking during

the first two turns, escalates from broad hint to stronger clue, detects when the student is stuck, and updates session memory based on repeated confusion or incorrect guesses.

5.3 Text Backend

The text backend generates the tutoring response once the retriever and controller have produced the grounded context and pedagogical state. The primary text model is served through a Groq-hosted inference path using Llama 3.1 8B Instruct (Groq, 2025), while GPT-4o-mini is retained as a fallback (OpenAI, 2025b).

5.4 Multimodal Backend

The multimodal backend infers the likely structure or topic shown in an uploaded diagram, uses aliases and metadata cues to map that inference to a text-retrieval target, and supports grounded follow-up tutoring rather than direct image labeling. This path is primarily designed around a LLaVA-NeXT/LLaVA-1.6-style model, with GPT-4o retained as a stronger fallback.

5.5 Session Memory and Assessment

The memory is session-only rather than persistent across users. It stores weak topics, recent incorrect guesses, tutoring-turn state, and repeated confusion signals. The assessment module performs a lightweight mastery check after tutoring by comparing the student’s free-text explanation and optional application response against retrieved source content.

Because the architecture separates the knowledge layer from the tutoring controller, the system is reusable across subjects. Replacing the domain-specific text and image knowledge base should allow the same design to support tutoring in other subjects without changing the overall control flow.

6 Technical Stack and Model Plan

We use LangGraph as the orchestration layer because it is designed for stateful agent workflows and controlled execution graphs (LangChain, 2025). Chunks and user queries are embedded with all-MiniLM-L6-v2 and stored in ChromaDB with metadata filtering support (SentenceTransformers, 2025; Chroma, 2025). The text tutoring backend is Groq-hosted Llama 3.1 8B Instruct (Groq, 2025) with GPT-4o-mini fallback (OpenAI, 2025b). For diagram understanding we use a LLaVA-NeXT/LLaVA-1.6-style path, with GPT-4o

Component	Primary	Fallback / Note
Text tutor backend	Groq + Llama 3.1 8B Instruct	GPT-4o-mini
Multimodal backend	LLaVA-NeXT / LLaVA 1.6	GPT-4o
Embedding model	Mistral 7B all-MiniLM-L6-v2	local embedding path
Vector database	ChromaDB	persistent metadata-aware retrieval
Orchestration	LangGraph	LangChain integrations where needed

Table 3: Planned backend and model configuration.

Metric	Type	Target
Faithfulness (RAGAS)	Quantitative	≥ 0.90
Answer Relevance (RAGAS)	Quantitative	≥ 0.85
Context Recall	Quantitative	≥ 0.80
Context Precision	Quantitative	≥ 0.80
Socratic Purity	Qualitative	$= 1.0$
Structure ID Accuracy	Quantitative	≥ 0.80
Follow-up Relevance	Human (1–5)	≥ 4.0
Domain Swap Test	Binary	Pass

Table 4: Evaluation metrics summary.

retained as a stronger fallback for difficult images (Liu et al., 2023b; OpenAI, 2025a). Retrieval uses top-5 semantic search, metadata-aware reranking, optional merging of adjacent same-section chunks, and final prompting with the best 2–3 refined context units.

7 Evaluation Methodology

The evaluation plan combines the professor’s required criteria with additional technical metrics for retrieval quality, tutoring quality, and multimodal performance.

7.1 Groundedness and Retrieval Quality

For groundedness, we use RAGAS-based metrics such as faithfulness and answer relevance (Es et al., 2024; Ragas, 2025). For retrieval quality, let $R_k(q)$ be the set of top- k retrieved chunks for query q , and let $G(q)$ be the set of gold relevant chunks. We compute

$$P@k = \frac{|R_k(q) \cap G(q)|}{|R_k(q)|} \quad (1)$$

and

$$R@k = \frac{|R_k(q) \cap G(q)|}{|G(q)|}. \quad (2)$$

7.2 Socratic Purity and Leakage Control

We define leakage rate as

$$LR = \frac{N_{\text{leak}}}{N_{\text{dialogues}}}, \quad (3)$$

where N_{leak} is the number of dialogues in which the answer is revealed before turn 3. Socratic purity is then

$$SP = 1 - LR. \quad (4)$$

7.3 Multimodal Performance

For image-based tutoring, multimodal correctness is defined as

$$MC = \frac{N_{\text{correct}}}{N_{\text{image}}}, \quad (5)$$

where N_{correct} is the number of correct image-to-concept identifications.

7.4 Pedagogical Quality and Setup

Follow-up questions are scored on a 1–5 rubric, where 1 is irrelevant or generic and 5 is grounded, useful, and reasoning-oriented. Hints are judged by whether they guide the learner toward the correct concept without revealing it too early. The planned evaluation set includes 8 text tutoring conversations, 6 image-based tutoring examples, and 10 blind test questions. Generalizability is tested by replacing the current domain knowledge base with another subject while keeping the architecture fixed.

8 Computational Environment

The project will primarily use Google Colab for development and experimentation, with local development for preprocessing, data organization, and lightweight prototyping. CCR remains an optional extension if larger experiments become necessary.

The expected setup includes Python for preprocessing, orchestration, and evaluation; automated PDF ingestion and chunk generation; local all-MiniLM-L6-v2 embeddings; ChromaDB for retrieval; Groq-hosted Llama 3.1 8B Instruct for text tutoring; LLaVA-NeXT for multimodal reasoning; and LangGraph for orchestration and session-state control. The compute demand for Milestone 1 is modest because the current work focuses on knowledge-base construction, model selection, architecture design, and evaluation planning rather than large-scale model training.

9 Project Plan and Responsibilities

The implementation proceeds in five stages: knowledge-base construction, retrieval setup, tutoring logic, multimodal integration, and final evaluation.

Phase	Task Split
Phase 1	Vidhyadhari: literature review support, text data download, section extraction, cleaning, chunk generation, and text knowledge-base evidence. Richie: image collection, image metadata creation, multimodal data organization, and backend/model comparison support. Shared: architecture design, problem framing, evaluation plan, report writing, and presentation preparation.
Phase 2	Vidhyadhari: tutoring-policy logic, masking rules, escalation design, stuck-student detection, and session-memory updates. Richie: embedding pipeline, ChromaDB setup, retrieval implementation, metadata-aware reranking, and baseline retrieval experiments. Shared: retrieval-generation integration, prompt refinement, and intermediate system testing.
Phase 3	Vidhyadhari: mastery-check flow, final tutoring behavior refinement, and domain-transfer prompt adjustments. Richie: multimodal pipeline refinement, blind diagram evaluation, quantitative evaluation scripts, and domain-swap test support. Shared: full evaluation, baseline comparison, final report writing, and final presentation delivery.

Table 5: Phase-wise task split between team members.

10 Conclusion

Socratic-OT is designed as a grounded, multimodal, and pedagogically controlled tutor for anatomy and neuroscience learning in rehabilitation science. The current milestone already demonstrates concrete progress through the construction of a text knowledge base, an image knowledge base, and a neuron metadata example that connects diagram understanding to grounded tutoring. More broadly, the architecture is designed to remain reusable across subjects by replacing the underlying knowledge base.

References

Articulate Community. 2024. Create an interactive branching scenario template in storyline 360. <https://community.articulate.com/kb/storyline-tips-tricks/create-an-interactive-branching-scenario-template-in-storyline-360/1216770>. Accessed 2026-03-22.

BoodleBox. 2025a. Boodlebox | ai built for lifelong learning and work. <https://boodlebox.ai/>. Accessed 2026-03-22.

BoodleBox. 2025b. Creating custom bots. <https://boodlebox.ai/blog/support/creating-custom-bots/>. Accessed 2026-03-22.

Chroma. 2025. Chroma documentation: Introduction. <https://docs.trychroma.com/docs/overview/introduction>. Accessed 2026-03-22.

Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. *Ragas: Automated evaluation of retrieval augmented generation*. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.

Groq. 2025. Groq documentation. <https://console.groq.com/docs/overview>. Accessed 2026-03-22.

Khan Academy. 2025. Khanmigo for learners: Always-available tutor, powered by ai. <https://www.khanmigo.ai/learners>. Accessed 2026-03-22.

LangChain. 2025. Langgraph overview. <https://docs.langchain.com/oss/python/langgraph/overview>. Accessed 2026-03-22.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023a. *Visual instruction tuning*. *arXiv preprint arXiv:2304.08485*.

Haotian Liu and 1 others. 2023b. *Improved baselines with visual instruction tuning*. *arXiv preprint arXiv:2310.03744*.

Jiayu Liu, Zhenya Huang, and 1 others. 2024. *Socraticlm: Exploring socratic personalized teaching with large language models*. In *Advances in Neural Information Processing Systems*.

OpenAI. 2025a. Gpt-4o. <https://developers.openai.com/api/docs/models/gpt-4o>. Accessed 2026-03-22.

OpenAI. 2025b. Gpt-4o mini. <https://developers.openai.com/api/docs/models/gpt-4o-mini>. Accessed 2026-03-22.

OpenStax. 2022. Anatomy and physiology 2e. <https://openstax.org/details/books/anatomy-and-physiology-2e>. Accessed 2026-03-22.

Ragas. 2025. Ragas documentation. <https://docs.ragas.io/en/stable/>. Accessed 2026-03-22.

Sentence-Transformers. 2025. all-minilm-l6-v2 model card. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>. Accessed 2026-03-22.